



Mustafa Sibai

Chief Technology Officer

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About

Engineering leader with 14 years of experience across game development, backend systems, and infrastructure. Currently serving as the CTO at a funded MMO game studio, leading a team of 13 engineers while owning full technical strategy, architecture, hiring, and cross-functional alignment with product, art, and design leadership.

Built large-scale multiplayer backend systems from scratch including real-time party, chat, matchmaking, phasing, and server

Experience

Chief Technology Officer

Feb 22 - Present

Couch Heroes

- Led and mentored a team of up to 13 engineers building an MMO in Unreal Engine 5.
- Defined the organization technology stack and helped create a three-year production roadmap with the help of the game producer, game director, and art director.
- Hired and interviewed engineers, game designers, and artists.
- Defined coding standards and architecture guidelines for the engineering team.
- Designed and built a suite of backend services to support millions of players, including registration, authentication, session management, account management, payment system, social features, chat system, auction house, in-game store, inventory systems, dungeon finder, party system, analytics, and much more.
- Designed and developed 30+ microservices in Go, TypeScript, Node.js/Bun, Express, MongoDB, Redis, WebSocket, and TCP sockets.
- Built a real-time multithreaded party service in Go using raw TCP sockets with a custom protocol to serialise/deserialise data between Unreal Engine servers and the service. The service handles player invites, kicks, role updates, and more.
- Designed and developed a dungeon finder service written in Go and raw TCP sockets. The service supports role-based matchmaking for groups of five with tank, healer, and DPS roles. Matched players are moved to a dedicated server instance, and if the instance is full, a new one is automatically spun up.
- Built an instancing system for Unreal Engine using Kubernetes to manage world zones, dungeon instances, and player housing. Automatically spins up new server instances on demand and tears them down when empty, moving players between instances based on load, party invites, or zone transitions.
- Built a custom Kubernetes Horizontal Pod Autoscaler in TypeScript/Node.js to scale Unreal Engine process pods based on custom metrics such as player count.
- Built a real-time chat service in Go using WebSockets with a custom protocol to serialise/deserialise data between Unreal Engine servers and the service. The service supports proximity, private, guild, and zone chat. The service includes a profanity filter.
- Built a real-time notification system supporting in-game mail, friend requests, duel invites, achievement unlocks, and system announcements.
- Designed and developed a guild service in TypeScript, Bun, and MongoDB with support for creating guilds, inviting and kicking members, and general guild info management.
- Designed and developed a leaderboard service written in TypeScript, Express, Node.js/Bun and MongoDB as a standalone service to track and rank player race completion times.
- Designed and implemented a game launcher architecture using Unreal Engine, TypeScript, Node.js, Express, and MongoDB with support for patching, file hashing, corrupted

instanting. Designed and implemented cloud infrastructure, CI/CD pipelines, and microservices architecture supporting live game operations. Previously founded and led a game development studio, shipping titles across PC, console, mobile, and web.

Author of multiple custom game engines in C++ with DirectX 11 and OpenGL. Former university lecturer in game development. Passionate about building and mentoring high performing engineering teams and delivering technically ambitious products.

Education

Essex University Online
MSc Computer Science
2022 - 2024

Emirates Aviation University
HND Computer Science
2010 - 2012

Emirates Aviation University
NC Aerospace Engineering
2009 - 2010

Technical Skills

Game Development	14 yrs
Frontend Development	8 yrs
Backend Development	8 yrs
Engine Development	8 yrs
Network Infrastructure	4 yrs
Server Administration	4 yrs
LLM Infrastructure	1 yr
AI Development	1 yr
Native Mobile Development	1 yr
OS / Kernel Development	1 yr
C#	12 yrs
C++	8 yrs
JavaScript	8 yrs
TypeScript	8 yrs
Python	4 yrs
Go	2 yrs
Swift	1 yr
Rust	1 yr
x86 Assembly	1 yr
llama.cpp	1 yr
Ollama	1 yr
Open WebUI	1 yr

file repair, and leveraged Azure's built-in CDN for asset delivery.

- Designed and implemented multiple web applications using React, Redux, Redux Saga, and Tailwind CSS.
- Designed and developed the organization's CI/CD pipelines in Microsoft Azure for building, testing, and deployment.
- Containerized all services using Docker and scaled them with Kubernetes.
- Set up and managed a Perforce server for the entire organization on a Linux VM in Azure.
- Secured the organization servers behind a self-hosted WireGuard VPN with a private DNS using Nginx reverse proxy.
- Designed and implemented rate limiting and abuse prevention across all services.
- Designed and implemented a backup and disaster recovery strategy for all servers and databases.
- Designed and implemented an analytics and telemetry service using ClickHouse to track player behavior, session data, and errors in real time.
- Monitored servers using Prometheus, Grafana, and Uptime Kuma with Slack and email alerts.
- Managed 40+ Git repositories across Nightly, Dev, Stage, Prod, and PBE environments.
- Created and maintained Onboarding, Dev, DevOps, and Security handbooks.
- Conducted code reviews, performance reviews, and workshops on Git, Docker, Kubernetes, and CI/CD for a 40-person team.
- Managed the organization's cloud infrastructure as code using Terraform, including provisioning and configuring all VMs, networking, and services on Microsoft Azure.
- Built internal tooling to manage, update, and monitor VMs across the organization's infrastructure.

Games Lecturer

Jun 17 – Aug 25

SAE Institute Dubai

- Taught the Bachelor's program in video game programming, including the following topics.
- Unity Engine with C#, Unreal Engine 5 with C++ and Blueprints.
- AI, including State Machine, A* Pathfinding, Behaviour Trees, GOAP, and Vision Cones.
- Multiplayer using TCP protocol in C# Sockets.
- 3D game math, including Vectors and Matrices.
- Computer Graphics, Raytracing, HLSL, GLSL, and CG Shaders.
- Project management using Agile methodology.

Founder - CTO

Oct 19 – Feb 22

Blue Monocle Software

- Started a game, animation, and App development studio.
- Developed and produced Apps and games in C++, C#, Unity, JavaScript, React, React Native, Node.js, and Gatsby.
- Recruited and managed programmers, game designers, artists, and audio engineers.

Senior React Native Developer

Jun 21 – Jan 22

Alshaya Group

- Optimized the performance of Victoria Secret app.

LangChain	1 yr
ChromaDB	1 yr
RAG	1 yr
React / React Native	6 yrs
Redux / Redux-Saga	4 yrs
Three.js	4 yrs
Next.js	1 yr
Vue.js	1 yr
Flutter	1 yr
Django / Flask	1 yr
Node.js / Bun	10 yrs
Express.js	10 yrs
JWT / OAuth 2.0	4 yrs
Kafka / RabbitMQ	2 yrs
MongoDB	8 yrs
Redis	6 yrs
PostgreSQL	4 yrs
GraphQL	2 yrs
Bash	4 yrs
PowerShell	4 yrs
Unity Engine	10 yrs
Unreal Engine	4 yrs
FMOD	2 yrs
DirectX 11	8 yrs
OpenGL 4.5	8 yrs
HLSL	8 yrs
GLSL	8 yrs
Emscripten	4 yrs
DirectX 12	1 yr
Vulkan	1 yr
Metal	1 yr
Socket Programming	8 yrs
WebSocket	6 yrs
Mirror / PUN	4 yrs
WebRTC	2 yrs
gRPC	2 yrs
Microsoft Azure	8 yrs
Google Cloud	5 yrs
Firebase	2 yrs
AWS	1 yr
Git	10 yrs
Perforce	2 yrs
Azure DevOps CI/CD	8 yrs
Docker	8 yrs
Nginx	8 yrs
Kubernetes	6 yrs
Cloudflare	4 yrs
Proxmox	4 yrs
Github Actions	4 yrs
Terraform	2 yrs
Grafana / Prometheus	1 yr

Founding Partner - CTO

Oct 19 - Jan 21

Beautify

- Defined the company technology and hired a team to create a mobile-based booking and E-commerce platform on both Android and IOS.
- Managed the entire app development cycle from start to finish while also helping develop the application with the programming and designing team.

React Native Developer

Sep 20 - Nov 20

Emaar Entertainment LLC

- Optimized the App by reducing asset sizes and converting them to SVG format.
- Added app localisation and support for RTL (Right-to-Left) languages and translated all text into Arabic.

CTO

Apr 19 - Jul 19

Coded Minds

- Defined the company technology and hired a team to create the web-based / Android and IOS education platform.
- Recruited and led a team in designing a STEM-based curriculum and created VR/AR-based tools and educational games.

Lead Game Developer

Feb 18 - Aug 18

SRH Studios

- Recruited and led a team of game programmers and artists to develop mobile games, VR and AR content using Unity. Shipped titles include River Run (3D endless runner) and Jump AR (3D AR game), both released on iOS and Android.

Game Developer

Sep 12 - Oct 17

Vault 16 Software

- Designed, implemented, and published multiple games for various clients in C++ and C# on Android and iOS, including Race Start 2 – a 2D singleplayer/multiplayer platformer built in C# and Unity using the Photon framework for real-time networking, The Exiled – a procedurally generated 2D pixel art dungeon game built in C# and XNA, and Heroes of Newcastle – a 2D tower defense game built in C# and XNA with a custom game engine, using UDP sockets for gameplay networking and TCP for chat.

Network Engineer

Mar 17 - Jun 17

Digital Tree Software

- Designed and implemented server/client-based network code using the Unity game engine.

Commercial Projects

Couch Heroes

Feb 22 - Present

Unreal Engine 5 | [Windows](#)

Agile / Scrum
Jira
ClickUp
Asana

8 yrs
8 yrs
8 yrs
2 yrs

Victoria Secret

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

H&M

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

FootLocker

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

American Eagle

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

Bugged Dungeon

Jan 21 - Feb 22

C# | Unity • [Steam](#) | [Apple Store](#) | [Google Play Store](#)

Reel Cinema App

Sep 20 - Nov 20

JavaScript | React Native • [Apple Store](#) | [Google Play Store](#)

Personal Engines

4E4F5641 Engine

Dec 25 - Present

C++ | DirectX 11 | Windows • [GitHub](#)

- Building a custom MMO game engine and backend from scratch in C++ and DirectX 11, featuring a multi-layer terrain system with splat map texture blending, foliage and vegetation rendering, post-processing pipeline, skybox, material system, and GLTF 3D model loading.
- Developed accompanying MMO backend services in Go including authentication server, registration server, quest system, and items server with CI/CD pipeline via Gitea.

Blue Flame Engine

Mar 16 - Mar 18

C++ | OpenGL 4.5 | DirectX 11 | Windows | Linux
Android | Web • [GitHub](#)

- Developed an Entity/Component-based game engine that supports 2D and 3D rendering, Forward rendering, sprite batch rendering, Post-processing, Phong shading, point, spot, and directional light system, 3D math library, 2D physics engine, A* pathfinding, networking, and a C++ scripting system.

Vault Engine

Mar 14 - Apr 15

C# | XNA | Windows • [GitHub](#)

- Developed a 2D tile-based game engine which supports frustum culling, multithreaded A* pathfinding using a custom job system, and networking with server/client-based architecture using TCP and UDP protocol written in Winsocket2.

Cloud Infrastructure

Private Cloud Infrastructure

2022 – Present

Proxmox | Docker | Unreal Horde | WireGuard |
Ubiquiti | Nginx | Linux

- Designed and operate a private cloud infrastructure featuring an 8-node Proxmox cluster, 25Gbps Ubiquiti networking with VLAN segmentation and enterprise WiFi, WireGuard VPN, and 500TB+ storage across 8 servers.
- Implemented Unreal Engine Horde distributed build system across multiple servers, reducing Unreal Engine compile times for personal game development projects.
- Built CI/CD pipelines with self hosted Gitea and agents for personal project automation, following the same engineering standards applied in production environments.
- Designed and implemented a full backup and disaster recovery strategy using Kopia and URBackup across all services and data.
- Deployed and maintain a fully self hosted service stack via Nginx Proxy Manager, replacing all cloud dependencies including git hosting, email, storage, communication, and collaboration infrastructure.